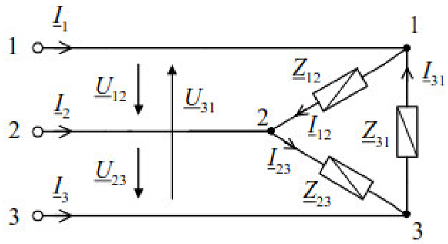
Mehrphasensysteme

Dreieck-Schaltung



Figur 4-27: Dreieck-Schaltung (der Last)

Mit $\underline{I} = \underline{U} \cdot \underline{Y}$ gelten:

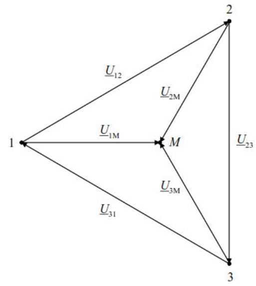
$$\underline{I}_{12} = \underline{U}_{12} \cdot \underline{Y}_{12} \qquad \underline{I}_1 = \underline{I}_{12} - \underline{I}_{31}$$

$$\underline{I}_{23} = \underline{U}_{23} \cdot \underline{Y}_{23} \qquad \underline{I}_2 = \underline{I}_{23} - \underline{I}_{12}$$

$$\underline{I}_{31} = \underline{U}_{31} \cdot \underline{Y}_{31} \qquad \underline{I}_3 = \underline{I}_{31} - \underline{I}_{23}$$

$$\underline{S} = \underline{U}_{12} \cdot \underline{I}_{12}^{\,*} + \underline{U}_{23} \cdot \underline{I}_{23}^{\,*} + \underline{U}_{31} \cdot \underline{I}_{31}^{\,*}$$

Zeigerdiagramm Drehstromsystem:



$\underline{U}_{1M} = U\angle 0°$

$\underline{U}_{2M} = U\angle -120°$

$\underline{U}_{3M} = U\angle -240°$

Man sieht leicht, dass

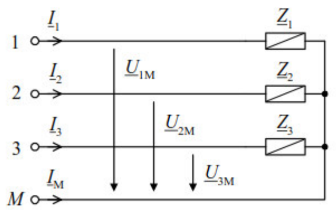
$$\boxed{\underline{U}_{1M} + \underline{U}_{2M} + \underline{U}_{3M} = 0}$$

und

$$\boxed{\underline{U}_{12} + \underline{U}_{23} + \underline{U}_{31} = 0}$$

Der Betrag der verketteten Spannung ist um sqrt(3) grösser als der Betrag der Phasenspannung

Stern-Schaltung mit Mittelpunktsleiter

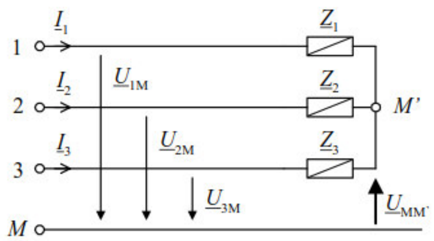


Figur 4-28: Stern-Schaltung (der Last) mit Mittelpunktsleiter

$$\underline{I}_1 = \underline{U}_{1M}/\underline{Z}_1, \quad \underline{I}_2 = \underline{U}_{2M}/\underline{Z}_2, \quad \underline{I}_3 = \underline{U}_{3M}/\underline{Z}_3$$

$$\underline{I}_1 + \underline{I}_2 + \underline{I}_3 + \underline{I}_M = 0$$

Stern-Schaltung ohne Mittelpunktsleiter



Figur 4-29: Stern-Schaltung (der Last) ohne Mittelpunktsleiter

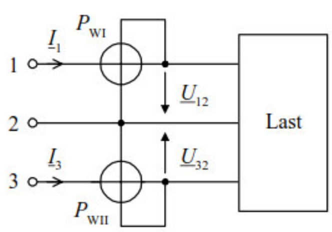
$$\underline{I}_1 = \underline{U}_{1M'}/\underline{Z}_1, \quad \underline{I}_2 = \underline{U}_{2M'}/\underline{Z}_2, \quad \underline{I}_3 = \underline{U}_{3M'}/\underline{Z}_3$$

$$\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0$$

$$\underline{U}_{1M'} \cdot \underline{Y}_1 + \underline{U}_{2M'} \cdot \underline{Y}_2 + \underline{U}_{3M'} \cdot \underline{Y}_3 = 0$$

$$\underline{U}_{MM'} = -\frac{\underline{U}_{1M} \cdot \underline{Y}_1 + \underline{U}_{2M} \cdot \underline{Y}_2 + \underline{U}_{3M} \cdot \underline{Y}_3}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3}$$

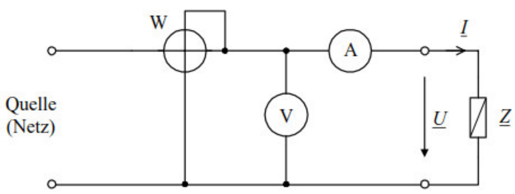
Leistungsmessung (Wattmeter)



P_{WI}, P_{WII}: Anzeigen der Wattmeter

P: Gesamte Wirkleistung

$$\boxed{P = P_{WI} + P_{WII}}$$

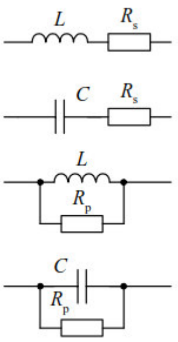


Merksätze:

- Wattmeter misst die Wirkleistung P
- Es braucht immer n-1 Wattmeter (N: Anzahl Leiter)
- Zur Leistungsmessung müssen Klemmenpaare gebildet werden

Gütefaktor Q

$$Q = 2\pi \cdot \frac{\text{maximal gespeicherte Energie}}{\text{verheizte Energie in einer Periode}}$$



$$\boxed{Q = \frac{\omega_0 L}{R_s}}$$

$$\boxed{Q = \frac{1}{\omega_0 C R_s}}$$

$$\boxed{Q = \frac{R_p}{\omega_0 L}}$$

$$\boxed{Q = \omega_0 C R_p}$$

Verlustfaktor:

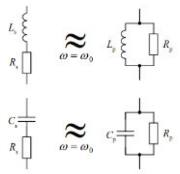
$$\tan(\delta) = \frac{1}{Q}$$

Q aus Phasensteilheit:

$$Q = \frac{1}{2} \cdot \omega_0 \cdot \left|\frac{d\varphi}{d\omega}\right|_{\omega=\omega_0} = \frac{1}{2} \cdot f_0 \cdot \left|\frac{d\varphi}{df}\right|_{f=f_0}$$

$$\frac{1}{Q_{tot}} = \frac{1}{Q_L} + \frac{1}{Q_C} \quad \text{oder} \quad Q_{tot} = \frac{Q_L \cdot Q_C}{Q_L + Q_C}$$

Serie-Parallel-Umformung



Für Q > 10 gilt:

$$L_s = L_p$$
$$C_s = C_p$$
$$R_p = R_s \cdot Q^2$$
$$R_s = R_p / Q^2$$

Für Q < 10 gilt:

$$R_s = \frac{R_p}{1 + \left[\frac{R_p}{\omega_0 L_p}\right]^2} = \frac{R_p}{1+Q^2}$$

$$R_p = R_s \cdot \left[1 + \left[\frac{\omega_0 L_s}{R_s}\right]^2\right] = R_s \cdot (1+Q^2)$$

$$R_s = \frac{R_p}{1 + (\omega_0 C_p R_p)^2} = \frac{R_p}{1+Q^2}$$

$$R_p = R_s \cdot \left[1 + \frac{1}{(\omega_0 C_s R_s)^2}\right] = R_s \cdot [1+Q^2]$$

$$L_s = \frac{L_p}{1 + \left[\frac{\omega_0 L_p}{R_p}\right]^2} = \frac{L_p}{1+1/Q^2}$$

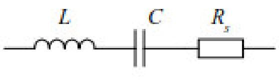
$$L_p = L_s \cdot \left[1 + \left[\frac{R_s}{\omega_0 L_s}\right]^2\right] = L_s \cdot (1+1/Q^2)$$

$$C_s = C_p \cdot \left[1 + \frac{1}{(\omega_0 C_p R_p)^2}\right] = C_p \cdot (1+1/Q^2)$$

$$C_p = \frac{C_s}{1 + (\omega_0 C_s R_s)^2} = \frac{C_s}{1+1/Q^2}$$

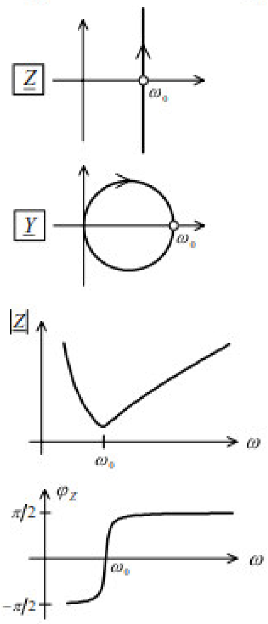
mit: $Q = \frac{R_p}{\omega_0 L_p} = \frac{\omega_0 L_s}{R_s}$ oder: $Q = \omega_0 C_p R_p = \frac{1}{\omega_0 C_s R_s}$

Serieschwingkreis

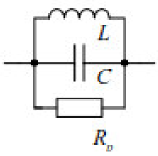


$$Q = \frac{1}{R_s}\sqrt{\frac{L}{C}}$$

$$\underline{Z} = R_s \cdot \left[1 + j \cdot Q \cdot \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right] = R_s \cdot [1 + j \cdot Q \cdot \upsilon]$$

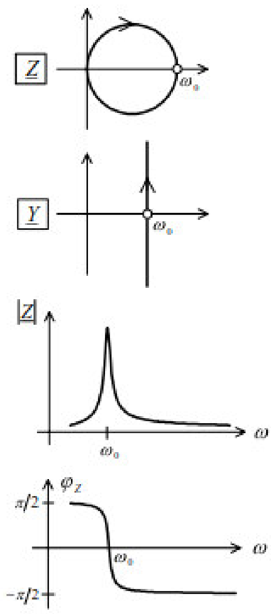


Parallelschwingkreis



$$Q = R_p\sqrt{\frac{C}{L}}$$

$$\underline{Y} = \frac{1}{R_p} \cdot \left[1 + j \cdot Q \cdot \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right] = \frac{1}{R_p} \cdot [1 + j \cdot Q \cdot \upsilon]$$



Resonanzfrequenz: $\omega_0 = \frac{1}{\sqrt{LC}}$

Verstimmung: $\upsilon = \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}$

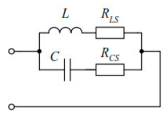
Untere Grenzfrequenz: $\omega_1 = \omega_0 \cdot \left[\sqrt{1 + \frac{1}{4Q^2}} - \frac{1}{2Q}\right] \approx \omega_0 \cdot \left[1 - \frac{1}{2Q}\right]$

Obere Grenzfrequenz: $\omega_2 = \omega_0 \cdot \left[\sqrt{1 + \frac{1}{4Q^2}} + \frac{1}{2Q}\right] \approx \omega_0 \cdot \left[1 + \frac{1}{2Q}\right]$

„Mittenfrequenz“: $\omega_0 = \sqrt{\omega_1 \cdot \omega_2} \approx \frac{1}{2}(\omega_1 + \omega_2)$

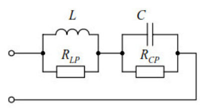
Bandbreite: $\Delta\omega = \omega_2 - \omega_1 = \omega_0 / Q$

Verlustabhängige Resonanzfrequenz



$$\boxed{\omega_r = \omega_0 \cdot \sqrt{\frac{1}{1 - 1/Q_C^2}} \cdot \sqrt{1 - 1/Q_L^2}}$$

$$Q_C = \frac{1}{R_{CS}} \cdot \sqrt{\frac{L}{C}}, \quad Q_L = \frac{1}{R_{LS}} \cdot \sqrt{\frac{L}{C}}$$



$$\boxed{\omega_r = \omega_0 \cdot \sqrt{\frac{1}{1 - 1/Q_L^2}} \cdot \sqrt{1 - 1/Q_C^2}}$$

$$Q_C = R_{CP} \cdot \sqrt{\frac{C}{L}}, \quad Q_L = R_{LP} \cdot \sqrt{\frac{C}{L}}$$

Achtung:

- Diese Formeln nur verwenden wenn die Resonanzfrequenz vom Verlustwiderstand (Q) abhängig ist!
- D.h. der Widerstand kommt in der Formel des Imaginärteils von $\underline{Z}$ bzw. $\underline{Y}$ vor
- **R → Im($\underline{Z}$)** bzw. **Im($\underline{Y}$)**

Betrag und Phase vom Serieschwingkreis:

$$|\underline{Z}| = Z = R_s \cdot \sqrt{1 + (Q \cdot \upsilon)^2}$$

$$\angle \underline{Z} = \varphi_Z = \arctan(Q \cdot \upsilon) = \arctan\left[Q \cdot \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]$$

Betrag und Phase vom Parallelschwingkreis:

$$|\underline{Y}| = Y = \frac{1}{R_p} \cdot \sqrt{1 + (Q \cdot \upsilon)^2} \Rightarrow Z = \frac{R_p}{\sqrt{1 + (Q \cdot \upsilon)^2}}$$

$$\angle \underline{Y} = \varphi_Y = \arctan(Q \cdot \upsilon) = \arctan\left[Q \cdot \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]$$